

Francis Anthony Konikkara

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Summary

Security-focused graduate student with 2+ years of hands-on experience across penetration testing, vulnerability assessment, cloud security, and secure software development. Former Deloitte Analyst with exposure to enterprise systems, ML-assisted security workflows, and DevSecOps practices. CEH certified, CTF competitor, and strong collaborator seeking Security Engineering and Offensive Security internships.

Technical Skills

Penetration Testing, Vulnerability Assessment, Web Security, API Security, Threat Modeling, Incident Response, SIEM (Splunk, ELK), MITRE ATT&CK, OWASP Top 10, Nessus, Metasploit, Burp Suite, Wireshark, Nmap, OSINT, AWS, Azure, GCP, Docker, Kubernetes, Terraform, CI/CD Security, Zero Trust, Linux Hardening, Python, Go, JavaScript, TypeScript, Bash, Power-Shell, SQL, ReactJS, Django, Flask, Node.js, SAP Analytics Cloud, Power BI, Tableau, Git, GitHub

Experience

Deloitte Touche Tomatsu LLP: , *Analyst*, Oct 2023 - Jul 2025

- Supported security-adjacent analytics and machine learning workflows using Python and JavaScript
- Assisted in secure data handling, model evaluation, and feedback pipelines for ML systems
- Worked across cloud platforms (AWS, Azure) supporting enterprise-scale analytics environments
- Coordinated deliverables across cross-functional teams while maintaining security and compliance requirements

Zedex Info Pvt. Ltd: , *Full Stack Development Intern*, Jul 2022 - Apr 2023

- Designed and deployed a secure web-based ERP system using React, Node.js, and GraphQL
- Implemented authentication, authorization, API security controls, and secure logging mechanisms
- Integrated third-party APIs while enforcing validation and error-handling best practices

CybersmithSecure Pvt Ltd: , *Information Security Analyst Intern*, Apr 2021 - Aug 2021

- Conducted vulnerability assessments and penetration testing using Nessus, Burp Suite, and Metasploit
- Identified web, network, and configuration vulnerabilities and supported remediation planning
- Authored detailed technical reports and collaborated with senior analysts on threat categorization

Education

University of Maryland: College Park, MD, *Master of Engineering in Cybersecurity*, 2025 – 2027

University of Mumbai: Mumbai, *Bachelor of Engineering in Computer Engineering*, 2019 – 2023

Key Security Projects

Secure CI/CD Pipeline (DevSecOps): Designed and enforced a compliance-ready CI/CD pipeline with embedded SAST, secrets scanning, and container vulnerability scanning. Implemented policy-as-code security gates and generated audit-grade evidence aligned with SOC 2 and ISO/IEC 27001 controls.

SIEM & Detection Engineering Lab: Built a SIEM platform using the ELK stack to ingest, normalize, and analyze security logs. Developed detection rules mapped to MITRE ATT&CK techniques, simulated attack scenarios, and validated alerts to support SOC triage and incident response workflows.

Malware Analysis Lab: Performed static and dynamic analysis of Windows and Linux malware samples. Extracted IOCs, analyzed execution behavior, and documented findings using reverse engineering and sandboxing techniques.

Reinforcement Learning for Security: Supported reinforcement learning models (RLHF) using Python and JavaScript to improve security decision-making accuracy through human feedback and iterative model refinement using AWS Sagemaker.

Certifications & Activities

Certifications: **Certified Ethical Hacker (CEH)**, Practical Ethical Hacking (TCM Security), Cisco Packet Tracer, AI for Everyone

CTF Competitions: UMBC Nightwing CTF (3rd Place), Fword CTF, Cyber Apocalypse CTF; experience in reverse engineering and exploitation challenges

Leadership & Outreach: President, Agnel Cyber Cell; led cybersecurity workshops, awareness sessions, and technical training events; produced educational content reaching 22K+ viewers